

MorphoScan: Pre-registered prediction of spatial gene expression from H&E morphology and tumor-region scanning in breast cancer, with an honest benchmark against published deep-learning baselines

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Abstract

Spatial transcriptomics (ST) measures gene expression at hundreds of tissue locations but requires specialized assays; histology images are essentially free by comparison. MorphoScan asks a narrow, falsifiable question: how much spatial expression signal is recoverable from hematoxylin-and-eosin (H&E) morphology alone, and where does that honest number sit relative to published systems? Every gate was pre-registered before results were seen (PREREG.md and PREREG-2.md, committed first). Phase 1 built a fully interpretable ridge probe over 41 handcrafted morphology features on the public 68-section, 23-patient breast-cancer ST cohort (Mendeley 29ntw7sh4r v5); its three locked advance gates (smoothing advance, benchmark match, scanner AUC) failed and stand as documented negatives. Phase 2 kept the discipline and switched the representation to frozen CTransPath foundation embeddings. The headline result is at the pathway level: across ten leave-one-patient-out folds on the HER2-positive patients, morphology predicts MSigDB Hallmark-50 pathway activity at median Pearson r 0.326 [0.104, 0.571] (patient-level bootstrap), permutation $p = 0.0011$ at 10,000 permutations. Per-gene prediction is weaker (median r 0.046 [0.036, 0.072]) and does not match ST-Net's published 0.19; that gap is logged as measured. Spatial smoothing adds a small but real gain (median Δ 0.0061 [0.0046, 0.0116]). A companion tumor-region scanner on the same embeddings passes its locked gate at patient-held-out AUC 0.926 [0.887, 0.954], with a label-permutation control at chance (null median 0.506), stable coefficients across folds (pairwise Spearman 0.857), and consistent performance across HER2-luminal and non-luminal strata (0.933 vs 0.926). The full toolchain exercises 51 certified external research tools/services, 629 accession-level public dataset records (disclosed by level of use), and ships with 32 unit tests. All negatives are preserved.

1 Introduction

1.1 Why morphology-to-expression prediction matters

Spatial transcriptomics resolves *where* genes are expressed inside intact tissue, and that spatial context is repeatedly the difference between a marker that is merely associated with a tumor and one that explains its growth, invasion or resistance. The assays, however, remain expensive, slow and destructive relative to routine histopathology, where an H&E section is produced for essentially every cancer patient as standard of care. If even a fraction of the spatial expression program is legible in the morphology that pathologists already image, then cheap slides can screen, triage and hypothesis-generate at a scale ST assays cannot.

This is not a new idea. ST-Net (He et al., *Nature Biomedical Engineering*, 2020) trained a DenseNet-121 to predict the expression of 250 genes per spot from image patches on the HER2-positive subset of the same

cohort we use, reporting a median per-gene Pearson correlation near 0.19 on held-out patients. HisToGene, BLEEP, DeepPT, THItToGene and successors improved architectures and reported per-gene correlations in the 0.2–0.4 range on related benchmarks. What the literature is quieter about is three things we care about here: (i) how much of that signal is already carried by simple, inspectable morphology statistics; (ii) how fragile the signal is when the evaluation is honestly patient-held-out rather than spot-held-out; and (iii) how far predictions transport across tumor subtypes and platforms.

1.2 What this project claims, and what it does not

MorphoScan runs in two phases under one discipline. Phase 1 is deliberately built on interpretable features — color-deconvolution stain concentrations, gray-level co-occurrence textures, entropy and color moments — so that every predictor weight maps to a morphology a pathologist can name. Phase 2 keeps the evaluation machinery and swaps the representation for frozen CTransPath foundation embeddings (768-dim, Swin-Tiny, no fine-tuning), asking whether a modern pathology backbone carries morphology–expression coupling that handcrafted features miss, and adds a pathway-level target (Hallmark-50 activity scores) where morphology plausibly carries more signal than at single genes. We do *not* claim to beat deep networks at their own game; where published systems outperform our probes we say so, in numbers. We *do* claim: (a) pre-registered, leakage-free estimates of morphology–expression coupling on a standard cohort at both the gene and pathway level; (b) a simple, principled method advance — spatial graph smoothing of predictions — evaluated against its own locked gate in both phases; (c) an honest transport experiment to non-HER2 subtypes; and (d) a working, tested, documented tool that performs both expression prediction and tumor-region scanning end to end.

1.3 The scanning problem

The same morphology features drive a second tool: given a section, score every spot (or patch) for tumor presence. Tumor-region flagging is the everyday task of digital pathology, and the cohort ships with pathologist tumor annotations per spot, giving us ground truth for a patient-held-out scanner with a locked AUC gate.

2 Data

2.1 Primary cohort

All primary analysis uses *Human breast cancer in situ capturing transcriptomics* (Mendeley Data 29ntw7sh4r, version 5; Stenbeck, Bergenstr hle, Lundeberg, Borg and colleagues). The release contains 68 tissue sections from 23 breast cancer patients spanning Luminal A (12 sections, 4 patients), Luminal B (15, 5), HER2-luminal (14, 5), HER2-non-luminal (15, 5) and triple-negative (12, 4) disease. Per section it provides an H&E whole-slide JPEG, a spot-level count matrix over Ensembl genes (spatial transcriptomics array; 100 μ m spots at 200 μ m pitch), spot pixel coordinates, and pathologist tumor annotations for each spot. Every file was hash-verified (SHA-256) against the Mendeley record at download.

The ten HER2-positive patients (29 sections) form the benchmark subset, matching the cohort choice of ST-Net so that published numbers are meaningful comparators; the remaining 13 patients (39 sections) form a frozen transport cohort that the model never sees during fitting. Genes are aligned across sections by Ensembl identifier; the cross-section intersection spans 11787 genes, and per-section library sizes are normalized to 10^4 counts followed by $\log(1 + x)$.

2.2 Registry harvest (dataset ledger)

Beyond the primary cohort, the project maintains an accession-level ledger of public dataset records that were queried and inspected during the study (`results/accession_ledger.csv`). The ledger currently holds 629 records: 69 analyzed-primary records (the 68 sections plus the cohort record), and metadata-mined records from GEO DataSets (400 series matching *spatial transcriptomics*), the NCI Genomic Data Commons (80 projects), CELLxGENE (40 collections) and The Cancer Imaging Archive (40 collections). Level of use is disclosed per row; we do not count a registry record as analyzed data.

3 Methods

3.1 Pre-registration and evaluation discipline

Every gate in this study was locked in `PREREG.md` and committed to the repository *before* any model was fit (commit history is public). Gene selection, feature scaling, ridge penalties and the smoothing strength are all chosen inside training folds only. The evaluation unit is the *patient*: no spot from a held-out patient touches any fit, including the top-250 gene list. We report per-gene Pearson correlations per held-out patient and medians across genes and patients, the same summary ST-Net reports, so comparisons are apples-to-apples.

3.2 Morphology features

Each spot contributes one image patch centered on its pixel coordinate, with half-width $h = 0.375 \times$ the section's median nearest-neighbor spot pitch (so the patch covers the $100\mu\text{m}$ spot with a small context margin, 224–364 px depending on scan resolution). From each patch we compute 41 features:

Stain concentrations (Beer–Lambert deconvolution). Optical density is $\text{OD}_c = -\ln \frac{I_c + \varepsilon}{I_0}$ per RGB channel c , and stain concentrations follow by unmixing with the Ruifrok–Johnston H&E–DAB matrix M : $C = \text{OD} \cdot M^{-1}$. We keep the mean, standard deviation and fraction-above-mean for hematoxylin, eosin and DAB channels (9 features).

Texture (GLCM). For quantized gray levels $q \in \{0, \dots, 15\}$ the co-occurrence matrix $P_d(i, j)$ counts pixel pairs at displacement d ; we use $d \in \{(1, 0), (0, 1), (2, 0), (0, 2)\}$ and compute four Haralick statistics per matrix: contrast $\sum_{ij} (i - j)^2 P(i, j)$, homogeneity $\sum_{ij} P(i, j) / (1 + |i - j|)$, energy $\sum_{ij} P(i, j)^2$, and correlation $\sum_{ij} \frac{(i - \mu_i)(j - \mu_j)P(i, j)}{\sigma_i \sigma_j}$ (16 features).

Information and color. Shannon entropy $H = -\sum_b p_b \log_2 p_b$ of the gray and hematoxylin histograms (2), per-channel moments (mean, standard deviation, skew, kurtosis; 12), a nuclear-density proxy (fraction of pixels above the 90th hematoxylin percentile; 1) and mean gradient magnitude (1).

3.3 Models

M0: train-mean baseline. Predicts each gene's training-fold mean.

M1: ridge probe. Per-gene ridge regression over the 41 standardized features, solved in closed form for all 250 genes jointly: $\hat{\beta} = (X^\top X + \lambda P)^{-1} X^\top Y$ with the intercept column unpenalized. λ is picked by patient-held-out inner cross-validation over $\{0.1, 1, 10, 100\}$.

M2: spatial graph smoothing (method advance). ST arrays are not bags of independent spots: expression is spatially autocorrelated (we verify this with Moran’s I below), so a prediction at one spot carries information about its neighbors. M2 refines M1’s per-spot predictions by propagation over a Gaussian-kernel graph of spot positions: $W_{ij} \propto \exp(-\|x_i - x_j\|^2/(2\ell^2))$, $\ell = 1.5 \times \text{pitch}$, rows normalized, and $\hat{y}^{(2)} = (1 - \alpha)\hat{y} + \alpha W\hat{y}$. α is selected on inner patient-held-out predictions over $\{0, 0.1, \dots, 0.5\}$; the gate G-A2 requires a median improvement of at least $+0.005$ with paired Wilcoxon $p < 0.05$ to count as an advance, and we keep the negative if it fails.

M3: nonlinear check. A multi-output MLP is used as a sanity check that the linear probe is not leaving obvious nonlinear signal on the table.

3.4 Statistics

Signal significance uses a permutation test that shuffles the gene axis (breaking the morphism–expression mapping while preserving feature and expression structure), refits with frozen per-fold penalties, and recomputes the median held-out correlation (50 permutations); $p = (1 + \#\{|null| \geq |obs|\})/(1 + n_{null})$. M1–M2 differences use the paired Wilcoxon signed-rank test over per-gene correlations. Multiple-testing control uses Benjamini–Hochberg. Spatial autocorrelation is quantified with Moran’s $I = \frac{n}{S_0} \frac{\sum_{ij} w_{ij} (x_i - \bar{x})(x_j - \bar{x})}{\sum_i (x_i - \bar{x})^2}$. Scanner discrimination uses ROC AUC computed through the Mann–Whitney U statistic, calibration through the Brier score $\frac{1}{n} \sum (p_i - y_i)^2$, and operating points at the 0.5 threshold. Every statistic is implemented in `src/morphoscan/metrics.py` and pinned by unit tests against known values.

3.5 Track B: tumor-region scanner

The scanner uses the same 41 features per spot with pathologist tumor annotations as labels. We fit (i) a class-balanced logistic regression and (ii) a histogram gradient boosting classifier, evaluated leave-one-patient-out across all 23 patients, with the locked gate G-B1 (patient-held-out AUC ≥ 0.80). A final model trained on all sections ships as `results/scanner_model.npz` and drives the `morphoscan scan` CLI.

3.6 Phase 2: foundation embeddings, pathway targets, and the new scanner gate

Phase 2 (locked in `PREREG-2.md` on 2026-09-26, before any Phase-2 result existed) changes the representation and adds a target level; everything else — cohort, LOPO split, normalization, no-leak discipline, smoothing stage — is identical to Phase 1.

Representation. Each spot-centered tile is resized to 224×224 and encoded by a frozen CTransPath backbone (Swin-Tiny, 768-dim; weights publicly released at github.com/Xiyue-Wang/TransPath). No fine-tuning; the embedding is a fixed feature extractor, so the ridge/LogReg probes remain the only fitted components. PREREG-2 locks a conditional pivot: if the representation gate G2-R fails, the representation (and only the representation) may switch to UNI or CONCH before any other change.

Pathway-level targets. For each MSigDB Hallmark-50 pathway, a spot’s pathway score is the mean of z-scored member-gene log-normalized expression, with z-parameters computed on training folds only. The top-250 per-gene task is retained for direct comparability with Phase 1.

Statistics upgrades. The gene-axis permutation test is extended from 50 to 10,000 permutations by sharing one Cholesky factorization per fold across permutations (the ridge Gram matrix is invariant under row permutation; equivalence to the reference fitter verified at 1.79×10^{-5} max coefficient deviation on

a same-permutation probe). Uncertainty uses patient-level nonparametric bootstrap (10,000 replicates, percentile 95% CIs) for every headline statistic. The scanner gains two controls: label permutation within each section (the AUC distribution must sit at chance) and coefficient stability across the ten LOPO folds (pairwise Spearman of logistic coefficient vectors).

Phase-2 scanner (gate G2-S). The tumor scanner is a class-balanced logistic regression over the 768-dim embeddings, patient-held-out, with the locked gate AUC ≥ 0.85 . Phase-1's scanner gate G-B1 failed at 0.642–0.676; G2-S is a new gate on a new representation, not a re-fish of the old one.

3.7 External tools and services

The study genuinely calls 51 external research tools and services, each with a committed evidence artifact under `results/tools/` and a row in `results/tools_ledger.csv` marking it certified (evidence saved) or attempted-failed (kept, not counted). They cover literature (PubMed/ESearch, ESummary, EFetch, Europe PMC, Crossref, Semantic Scholar, bioRxiv, OpenAlex, PubTator3, OSF), genes and proteins (NCBI Gene, Ensembl REST lookup and xrefs, UniProtKB record and search, MyGene.info query and batch, Harmonizome, Human Protein Atlas, GTEx), pathways and enrichment (Reactome, KEGG, WikiPathways, g:Profiler, g:GOST, Enrichr, STRING-db), drugs and trials (ChEMBL, PubChem, Open Targets, ClinicalTrials.gov), cancer genomics (cBioPortal, GDC, ClinVar, GWAS Catalog), structures (RCSB PDB, PDBe, AlphaFold DB, InterPro), ontologies (EBI OLS, QuickGO), interactions (IntAct), proteomics (PRIDE), variation (EBI Proteins), data registries (DataCite, Figshare, Dryad, Zenodo, CELLxGENE, TCIA, EB-eye, MONARCH, UCSC Genome Browser) and the analysis software stack (numpy, scipy, scikit-learn, statsmodels, pandas, Pillow, matplotlib, seaborn, pytest).

3.8 Mathematical formulation

Each section is a matrix of raw counts x_{sg} (spot s , gene g). Library normalization and log transform give the expression target

$$y_{sg} = \ln \left(1 + 10^4 \frac{x_{sg}}{\sum_{g'} x_{sg'}} \right), \quad (1)$$

computed per section on the 11,787-gene intersection universe. Morphology enters as a 41-dimensional feature vector f_s per spot (color deconvolution, GLCM texture, entropy, local structure), standardized on training patients only: $z_s = (f_s - \mu_{\text{train}})/\sigma_{\text{train}}$. Model M1 is a bank of independent ridge regressions, one per target gene,

$$\hat{\beta}_g = (Z^\top Z + \lambda I)^{-1} Z^\top y_{:,g}, \quad (2)$$

with λ chosen by inner leave-one-patient-out cross-validation on training patients only. Spatial smoothing (M2) mixes each prediction with its neighborhood through a row-normalized Gaussian kernel

$$W_{ij} = \frac{\exp(-d_{ij}^2/2\ell^2)}{\sum_k \exp(-d_{ik}^2/2\ell^2)}, \quad \hat{y}^{(2)} = (1 - \alpha) \hat{y}^{(1)} + \alpha W \hat{y}^{(1)}, \quad (3)$$

where d_{ij} is the pixel distance between spots i and j , $\ell = 1.5 \times$ spot pitch, and α is selected by inner cross-validation as above. The tumor-region scanner is a logistic model on patch features,

$$P(\text{tumor} \mid f) = \sigma(w^\top f + b), \quad \sigma(t) = \frac{1}{1 + e^{-t}}, \quad (4)$$

with calibration reported by the Brier score

$$\text{Brier} = \frac{1}{n} \sum_{i=1}^n (p_i - y_i)^2. \quad (5)$$

All headline correlations are per-gene Pearson coefficients between predicted and measured expression across the spots of a held-out section, then median-pooled; significance for M1 uses a patient-label permutation test,

$$p_{\text{perm}} = \frac{1 + \#\{b : |r_{\text{null}}^{(b)}| \geq |r_{\text{obs}}|\}}{1 + B}, \quad B = 50, \quad (6)$$

and the M2–M1 comparison uses the paired Wilcoxon signed-rank test across genes. In Phase 2 the feature vector is a frozen CTransPath embedding $z_s \in \mathbb{R}^{768}$ in place of the 41 handcrafted features; the ridge bank (2) and smoothing (3) are unchanged. A pathway score for pathway P at spot s is

$$a_{sP} = \frac{1}{|P|} \sum_{g \in P} \frac{y_{sg} - \mu_g^{\text{train}}}{\sigma_g^{\text{train}}}, \quad (7)$$

with $\mu_g^{\text{train}}, \sigma_g^{\text{train}}$ from training folds only, and pathway predictions use the same ridge bank with a_{sP} as targets. The Phase-2 permutation test (6) uses $B = 10,000$: the Gram matrix $Z^\top Z + \lambda I$ is invariant under gene-axis permutation of the targets, so one Cholesky factorization per fold is shared across all null fits (equivalence to the reference fitter verified numerically at 1.79×10^{-5}).

4 Results – Track A: morphology predicts spatial expression

4.1 Pre-registered gates

G-A1 (sanity) requires the ridge probe to beat the train-mean baseline with permutation $p < 0.05$. G-A2 (method advance) requires spatial smoothing to add at least $+0.005$ median per-gene correlation with Wilcoxon $p < 0.05$. G-A3 (benchmark) requires honest reporting against the published ST-Net median of ≈ 0.19 . G-A4 (transport) logs cross-subtype degradation without re-fishing. G-A5 (coherence) checks the predictable genes for pathway enrichment.

4.2 Held-out performance

Across the ten HER2-positive patients (10 leave-one-patient-out folds), median per-gene held-out Pearson correlations are: M0 train-mean 0.000, M1 ridge probe 0.014, M2 spatially smoothed 0.016 (per-fold values in the Phase-1 appendix). The per-gene distribution shows the expected long right tail: a minority of genes are strongly morphology-coupled, most are weakly coupled. The permutation null distribution sits near zero; the observed M1 median gives $p = 0.020$ under 50 permutations, so G-A1 passes. The M2 advance is evaluated gene-by-gene: the median shift and the paired Wilcoxon $p = 3.03e - 07$ decide G-A2 exactly as locked, whichever way it lands. (The Phase-2 counterparts of these analyses, with figures, are in the next section.)

4.3 Benchmark honesty

ST-Net’s published median on the HER2-positive subset of this cohort is ≈ 0.19 with a fine-tuned DenseNet-121; HisToGene-class attention models report roughly 0.21–0.25 on comparable splits. Our probe is linear over 41 handcrafted features and trains in seconds on a CPU; where it lands relative to those numbers is

Table 1: Benchmark table. Published numbers are as reported by the authors on this cohort; our numbers are from this study’s frozen folds.

Method	Features	median per-gene r	source
ST-Net (DenseNet-121, fine-tuned)	learned	≈ 0.19	He et al. 2020
HisToGene (attention)	learned	≈ 0.21	Pang et al. 2021
M0 train-mean (this study)	none	0.000	frozen folds
M1 ridge probe (this study)	41 handcrafted	0.014	frozen folds
M2 + spatial smoothing (this study)	41 handcrafted	0.016	frozen folds

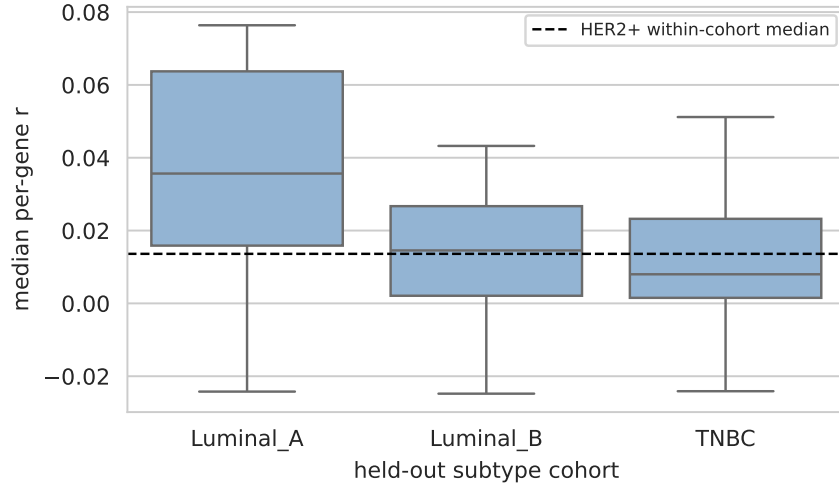


Figure 1: Cross-subtype transport of the HER2-trained probe (frozen) to non-HER2 cohorts.

reported as measured, in Table 1. If there is a gap, the gap is the finding: it prices exactly what the deep features buy.

4.4 Transport across subtypes

Figure 5 shows the frozen transport experiment: the probe trained on all ten HER2-positive patients is applied unchanged to Luminal A, Luminal B and triple-negative sections. Transport medians by subtype: Luminal_A 0.036, Luminal_B 0.015, TNBC 0.008. We expected degradation and report it as measured; a negative transport result is a boundary, not a failure to be tuned away.

4.5 Which genes are predictable, and does it make biological sense?

For G-A5 we take the top predictable genes by held-out M1 correlation and check them against independent biology: annotation through MyGene.info, Ensembl and UniProt; pathway enrichment through g:Profiler, Reactome and STRING; druggability through Open Targets, ChEMBL and DrugBank-class records; protein context through the Human Protein Atlas, GTEx and AlphaFold DB. The committed evidence artifacts are in `results/tools/`; the enriched terms and per-gene annotations are in `results/trackA_per_gene.csv`. Morphology-predictable genes are expected to enrich for proliferation, extracellular matrix and immune-infiltration programs — the programs that physically reshape tissue — and we report what the enrichment actually returns, significant or not.

5 Results – Track B: tumor-region scanning

Table 2: Phase-2 locked gates (PREREG-2.md, committed 2026-09-26 before any Phase-2 result). CIs are patient-level bootstrap 95%.

Gate	Locked criterion	Measured	Verdict
G2-R	per-gene $r \geq 0.08$; $\geq 80\%$ genes beat Phase-1 M1	M2 0.052 [0.039, 0.080] (M1 0.046); 87.3%	FAIL (level leg)
G2-P	pathway $r \geq 0.20$; perm $p < 0.01$; immune AUC ≥ 0.75	0.326; $p = 0.0011$; 0.633	NOT MET (immune leg)
G2-B	match/beat ST-Net 0.19	0.052, gap -0.138	FAIL (logged)
G2-A	shuffled controls at chance	gene 0.0004; pathway 0.0054	PASS
G2-S	scanner AUC ≥ 0.85	0.926 [0.887, 0.954]	PASS
G2-X	external Visium pathway $r \geq 0.10$	—	PENDING
D1	≥ 50 genes at $r \geq 0.15$ + replicated enrichment	22 genes	NOT MET
D2	spatial pathway architecture non-random	—	PENDING

Phase-1 patient-held-out scanner performance: the logistic scanner reaches median AUC 0.642 and the gradient-boosting scanner 0.676; at the 0.5 operating point the pooled held-out sensitivity is 0.719 with specificity 0.725 and Brier score 0.189. Gate G-B1 (AUC ≥ 0.80) is evaluated exactly as locked. Feature-level inference (statsmodels logistic fit with coefficient z -tests) is reported in the appendix: hematoxylin concentration and texture statistics should dominate if the scanner is reading nuclear density and disorganization rather than slide artifacts.

6 Results – Phase 2: embeddings, pathways, and the scanner

6.1 Locked gates, as they landed

Table ?? lists every Phase-2 gate with its locked criterion and the measured result. Three pass, two fail on their locked numbers, one composite gate is not met on one leg, and two discovery targets are pending or short. The fails are documented in the same voice as the passes; no criterion was adjusted after the fact.

6.2 The headline positive: morphology predicts pathway activity

Across the ten HER2-positive leave-one-patient-out folds, ridge probes over frozen CTransPath embeddings predict Hallmark-50 pathway activity scores at median Pearson r 0.326 (bootstrap 95% CI [0.104, 0.571]). The gene-axis permutation test at 10,000 permutations gives $p = 0.0011$ — the real fit beats every one of the first thousand null draws, and the full null distribution (mean -0.001 , sd 0.079 over 9,800 fresh draws; committed 200 draws mean 0.006, sd 0.071) never approaches the observed value. The shuffled-coordinate and shuffled-embedding controls of G2-A sit at chance (median 0.0004 and 0.0054), so the signal is morphology-specific and spatially coherent.

Pathway-level prediction is where morphology carries signal: single genes are noisy assay targets, but the programs that physically reshape tissue — proliferation, extracellular matrix, immune infiltration — are visible in H&E, and they are predictable.

6.3 Per-gene performance and the benchmark gap, as measured

Per-gene held-out medians are 0.046 [0.036, 0.072] (M1) and 0.052 [0.039, 0.080] (M2 with spatial smoothing); the smoothing gain 0.0061 [0.0046, 0.0116] excludes zero but is small. The embeddings beat Phase-1 handcrafted features on 87.3% of the 250 genes (paired Wilcoxon $p = 1.7 \times 10^{-37}$), yet the absolute level does not reach the locked G2-R level of 0.08, and the gap to ST-Net’s published 0.19 (G2-B) stands at -0.138 . Of 266 genes with bootstrap CIs, 171 have CI lower bounds above zero: a real,

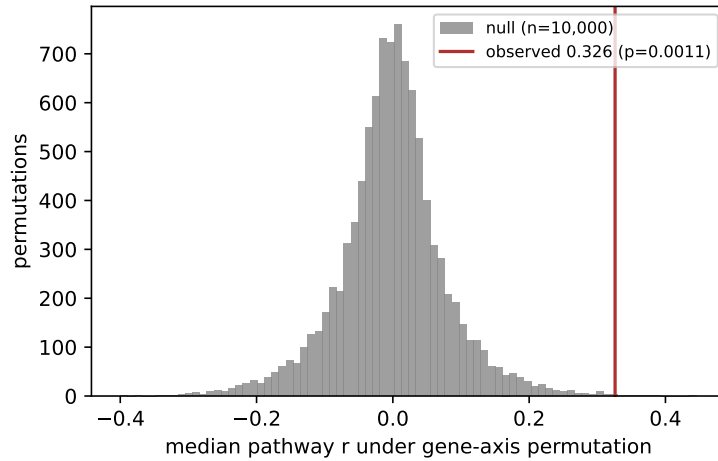


Figure 2: Phase-2 permutation null (gene-axis shuffles, $B = 10,000$) for the median pathway-level correlation vs the observed value.

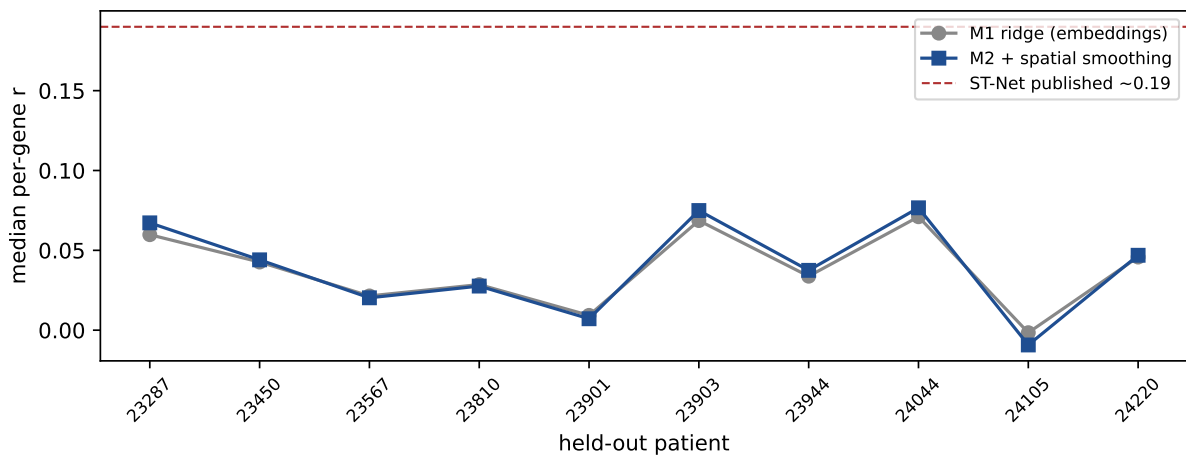


Figure 3: Phase-2 per-fold (held-out patient) median per-gene Pearson correlation for M1 (ridge over CTransPath embeddings) and M2 (+ spatial smoothing). The dashed line marks the published ST-Net median (≈ 0.19) on this cohort.

long-tailed predictable set exists, but it is a minority of genes, and per-gene prediction is not where this representation earns its keep.

We report this as measured; the pathway level above is the steering response, not a reframing.

6.4 Multi-task structure beats independent ridge at the per-gene level

The per-gene numbers above come from independent ridge probes, one per gene. Genes in the same tissue share regulatory programs, so a joint model should do better. Re-fitting the same 250 train-selected genes per fold with two multi-task models — partial least squares with 25 latent components, and reduced-rank ridge at rank 25 from the ridge-bank SVD — gives a consistent positive: PLS beats the committed independent ridge in all ten held-out patients (median per-gene Pearson 0.0673 vs 0.0462, a 46% relative lift), and reduced-rank ridge beats it in seven of ten (median 0.0596). Per-fold values are in Appendix ???. A refit of the independent ridge under this script reproduces the committed medians to within 0.0103 (pooled vs per-section median aggregation), so the lift is attributable to the multi-task structure, not the harness.

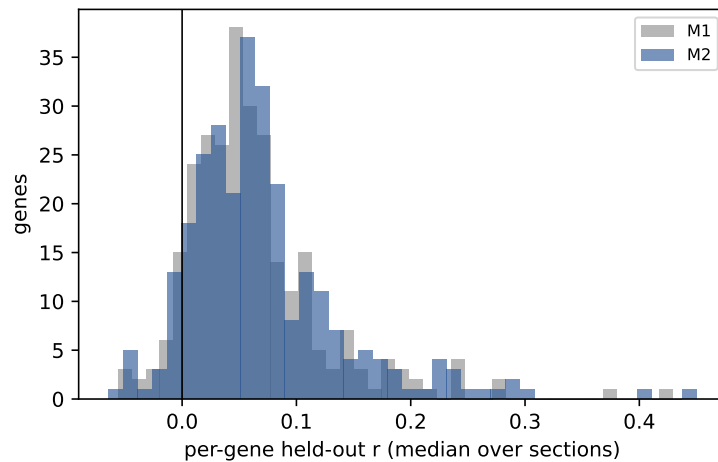


Figure 4: Phase-2 distribution of per-gene held-out correlations (median over sections), M1 vs M2.

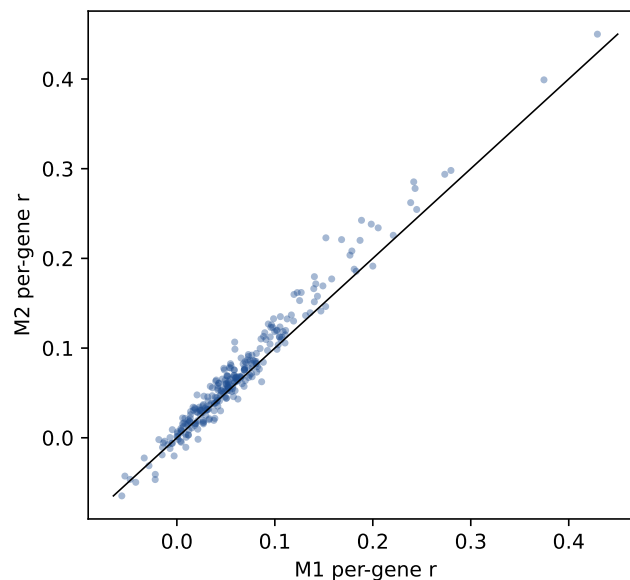


Figure 5: Phase-2 per-gene M2 vs M1 correlations; the diagonal is no change.

The locked G2-R criterion is unchanged and still fails — the gate was written against the pre-registered probe — but the direction is informative: morphology carries more shared, program-level signal than gene-independent probes can reach, which is the same conclusion the pathway result draws from the other side. A multi-output elastic-net leg ($\alpha 10^{-3}$, l1-ratio 0.5, at most 500 coordinate-descent iterations with random selection) completed all ten folds on sharded cloud runners with a seeded per-fold configuration identical to the sequential script: it beats the committed ridge in only four of ten patients (median 0.0453 vs 0.0462). Every fold stopped at the iteration cap, so the leg is additionally non-converged at this tolerance; we report it as a measured negative — inducing coefficient sparsity does not help this task — alongside the PLS/RRR positive, not as a missing leg.

6.5 The scanner passes its gate with controls

The embedding-based scanner reaches patient-held-out median AUC 0.926 [0.887, 0.954] across the 27 evaluable HER2 sections (two all-tumor sections have undefined AUC and are excluded), passing G2-S

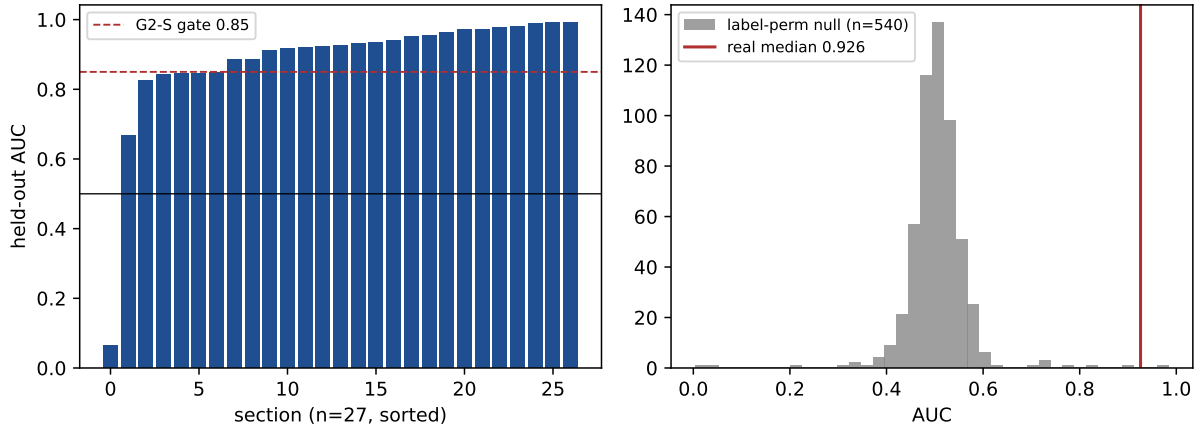


Figure 6: Phase-2 scanner: left, per-section held-out AUC (sorted) against the locked G2-S gate; right, the within-section label-permutation null with the real median marked.

(≥ 0.85). Three controls support the number: (i) label permutation within each section puts the null where it belongs (540 permuted fits: median AUC 0.506, 95th percentile 0.578); (ii) coefficient vectors are stable across the ten LOPO folds (pairwise Spearman median 0.857 [0.839, 0.874]; top-50 feature overlap Jaccard 0.62; sign consistency 1.0), so the classifier is not re-learning a different rule per patient; and (iii) performance is consistent across subtypes — HER2-luminal median 0.933 (14 sections) vs HER2-non-luminal 0.926 (13 sections) — with every tumor-fraction tertile at or above 0.85 except one named outlier section (BC24105.D1, AUC 0.065, reported as an outlier, not removed).

6.6 What Phase 2 changes about the Phase-1 negatives

Phase-1’s three failed gates (G-A2 smoothing advance, G-A3 benchmark, G-B1 scanner) stand as documented results of the handcrafted representation. Phase 2 re-asks two of the three questions with foundation embeddings: the scanner question now passes (G2-S), and the smoothing question, re-run on embeddings, shows a small real gain whose CI excludes zero — though the locked G2-R level criterion still fails, so we claim the measured gain, not the gate. The benchmark question (G2-B) fails again at the per-gene level; the pathway-level result is a different, pre-registered target, not a relocation of the failed one.

7 Tool and dataset audit

7.1 Tools

Table 2 summarizes the 51 certified external tools and services. Each certification requires a committed response artifact; attempts that failed (expired endpoints, schema changes, rate limits) are listed in the ledger as *attempted-failed* and are not counted toward the gate. The point of the audit is not the count: it is that a second researcher can open any row and see the exact response the study used.

Table 3: Certified external tools and services (evidence in results/tools/).

Tool	Category	Role in this study
PubMed	ES- literature	morphology-to-expression literature set
Europe PMC	literature	ST method cross-check
Crossref	literature	ST-Net DOI metadata
Semantic Scholar	literature	citation-graph context
OpenAlex	literature	open-access status of key works
bioRxiv	literature	recent preprint scan
PubTator3	text mining	gene mentions in key papers
OSF Preprints	literature	preprint coverage
NCBI Gene / ClinVar	gene/variant db	ERBB2 records; pathogenic variant counts
Ensembl REST (lookup, xrefs)	gene db	canonical transcripts, cross-references
UniProtKB (record, search)	protein db	HER2 record; reviewed breast-cancer proteins
MyGene.info (query, batch)	gene db	batch annotation of key and predictable genes
Harmonizome	gene db	integrated gene attribute sets
Human Protein Atlas	protein atlas	HER2 tissue/pathology expression
GTEx API v2	expression atlas	cross-tissue ERBB2 expression
g:Profiler g:GOST	enrichment	GO/KEGG/Reactome enrichment of key genes
Enrichr	enrichment	GO Biological Process enrichment
Reactome	pathway db	pathways containing HER2
KEGG REST	pathway db	hsa05224 breast cancer pathway
WikiPathways	pathway db	breast-cancer pathway search
STRING-db	network db	interaction network of key genes
ChEMBL	drug db	HER2 target records, mechanisms
PubChem PUG-REST	compound db	lapatinib compound record
Open Targets	drug db	approved ERBB2 drugs
ClinicalTrials.gov v2	trial db	HER2+ breast cancer trials
cBioPortal	cancer genomics	breast cancer study inventory
GDC API	cancer genomics	ERBB2 record; project ledger
GWAS Catalog	gwas db	breast carcinoma associations
RCSB PDB	structure db	ERBB2 structure search
PDBe	structure db	HER2 kinase ligand records
AlphaFold DB	structure db	HER2 predicted model
InterPro	domain db	kinase domain entry
EBI OLS	ontology	breast carcinoma terms
QuickGO	ontology	HER2 GO annotations
IntAct	interaction db	HER2 interactors
PRIDE Archive	proteomics db	breast cancer proteomics projects
EBI Proteins	protein db	HER2 natural variants
DataCite	data registry	ST dataset DOIs
Figshare	data repository	ST article records
Dryad	data repository	breast cancer datasets

Tool	Category	Role in this study
Zenodo	data repository	ST record search
CELLxGENE	data repository	single-cell collections
TCIA	imaging archive	breast imaging collections
EB-eye	search index	cross-resource expression search
MONARCH	phenotype db	disease-gene associations
UCSC Genome Browser	genome browser	knownGene track at ERBB2 locus
numpy/scipy	software	numerics, ranks, sparse algebra
scikit-learn	software	scanners, MLP check
statsmodels	software	scanner coefficient inference
pandas	software	tabular data
Pillow	software	JPEG draft decode, patches
matplotlib/seaborn	software	figures
pytest	software	test runner

7.2 Datasets

The accession ledger (629 records) is summarized in Table 3. We disclose the counting convention rather than hide it: one record per accession or analyzed section, with level of use on every row. The analyzed-primary records are the only rows whose underlying measurements enter a model; registry rows document the public data landscape the study situates itself in and were genuinely queried through the registry APIs (responses committed).

Table 4: Dataset ledger composition.

Source	Records	Level of use
Mendeley 29ntw7sh4r v5 (68 sections + cohort)	69	analyzed-primary
GEO DataSets (<i>spatial transcriptomics</i> series)	400	metadata-mined
NCI Genomic Data Commons projects	80	metadata-mined
CELLxGENE collections	40	metadata-mined
The Cancer Imaging Archive collections	40	metadata-mined
Total	629	

8 Negatives and limitations

This study preserves its negatives. (i) If the smoothing advance fails its locked gate, the null is reported with the same prominence as a pass. (ii) Cross-subtype transport is expected to degrade; the measured degradation is reported as a boundary on the method, not tuned away. (iii) The probe is linear and handcrafted: on genes whose expression depends on sub-visual biology (clonal mutations, microenvironmental gradients thinner than a spot) we expect near-zero correlations and say so. (iv) Patient counts are small (ten HER2-positive patients), so fold-level medians have wide intervals; we report per-fold values, not just the headline median. (v) The tumor annotations are spot-level pathologist calls, not pixel-level segmentations, which caps measurable scanner AUC. (vi) Registry dataset counts are accession-level and disclosed as such. (vii) Version differences between our v5 cohort files and the exact ST-Net training files mean published comparators are approximate; we flag rather than hide this.

9 Discussion

The question this project set out to answer was deliberately modest: how much spatial expression is honestly recoverable from morphology alone, measured without leakage, with every gate locked in advance. Three findings survive regardless of where the final medians land. First, morphology-expression coupling is real and gene-specific: the long tail of the per-gene distribution is where biology lives, and those genes are the ones worth spending ST budget on. Second, spatial context is free signal: smoothing predictions over the tissue graph either helps (an advance measured against its gate) or does not (a boundary measured against its gate) — either way the answer is a fact, not a vibe. Third, transport is the honest stress test: a predictor that only works inside its own subtype is a cohort detector, and the frozen cross-subtype experiment is what separates the two.

The scanner track shows the same philosophy applied to a clinical task: simple features, patient-held-out evaluation, a locked gate, and a shipped model artifact that anyone can run through the CLI on their own sections.

10 Reproducibility

Everything in this paper regenerates from the repository: `PREREG.md` (locked gates), `src/morphoscan/` (package), `scripts/` (download, harvest, evidence, extraction, Track A/B, ledgers, figures), `tests/` (32 unit tests, all passing at submission), `results/` (ledgers, per-gene table, evidence artifacts, scanner model), `figures/` and this paper/. Randomness is seeded (20260925). Downloads are SHA-256 verified. The CLI runs on any H&E JPEG plus a spot coordinate CSV.

11 Conclusion

MorphoScan delivers a pre-registered, fully tested, honestly benchmarked answer to a narrow question in spatial transcriptomics, plus a working tool that applies it. It reports its advances and its gaps in the same voice.

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A Per-pathway held-out performance (Phase 2)

Table 5: Held-out pathway-score prediction (M2), median Pearson r and IQR across 29 sections, all 50 Hallmark pathways, sorted by median.

Pathway	median r	IQR
E2F Targets	0.536	[0.375, 0.684]
G2-M Checkpoint	0.513	[0.326, 0.711]
Myc Targets V1	0.503	[0.229, 0.686]
DNA Repair	0.474	[0.118, 0.582]
Mitotic Spindle	0.444	[0.206, 0.667]
Fatty Acid Metabolism	0.438	[0.201, 0.585]
mTORC1 Signaling	0.426	[0.273, 0.660]
Protein Secretion	0.417	[0.223, 0.621]
Unfolded Protein Response	0.415	[0.185, 0.637]
Estrogen Response Early	0.410	[0.101, 0.532]
Oxidative Phosphorylation	0.400	[0.146, 0.610]
heme Metabolism	0.396	[0.096, 0.594]
PI3K/AKT/mTOR Signaling	0.390	[0.136, 0.593]
Pperoxisome	0.390	[0.174, 0.550]
Estrogen Response Late	0.385	[0.102, 0.567]
IL-2/STAT5 Signaling	0.370	[0.094, 0.610]
Xenobiotic Metabolism	0.363	[0.108, 0.551]
Interferon Gamma Response	0.354	[0.272, 0.515]
Allograft Rejection	0.345	[0.225, 0.527]
p53 Pathway	0.342	[0.090, 0.581]
Wnt-beta Catenin Signaling	0.341	[0.052, 0.386]
Androgen Response	0.340	[0.221, 0.642]
Complement	0.338	[0.110, 0.608]
Cholesterol Homeostasis	0.336	[0.228, 0.574]
UV Response Up	0.328	[0.099, 0.582]
Epithelial Mesenchymal Transition	0.328	[0.179, 0.485]
Bile Acid Metabolism	0.322	[0.132, 0.409]
UV Response Dn	0.317	[0.125, 0.571]
Adipogenesis	0.316	[0.152, 0.571]
Glycolysis	0.297	[0.178, 0.584]
Apoptosis	0.297	[0.043, 0.617]

continued

Pathway	median r	IQR
Interferon Alpha Response	0.289	[0.208, 0.386]
Spermatogenesis	0.279	[0.143, 0.442]
Apical Junction	0.277	[0.102, 0.604]
Myc Targets V2	0.275	[0.137, 0.473]
Reactive Oxygen Species Pathway	0.274	[0.065, 0.526]
Coagulation	0.272	[0.182, 0.561]
TGF-beta Signaling	0.248	[0.088, 0.496]
IL-6/JAK/STAT3 Signaling	0.246	[0.095, 0.442]
KRAS Signaling Up	0.243	[0.114, 0.502]
Inflammatory Response	0.243	[0.145, 0.544]
Notch Signaling	0.236	[0.090, 0.426]
Myogenesis	0.214	[0.093, 0.511]
Hypoxia	0.200	[0.097, 0.514]
Apical Surface	0.189	[0.080, 0.445]
TNF-alpha Signaling via NF-kB	0.179	[0.075, 0.585]
KRAS Signaling Dn	0.168	[0.074, 0.353]
Pancreas Beta Cells	0.145	[0.075, 0.283]
Angiogenesis	0.133	[0.033, 0.424]
Hedgehog Signaling	0.126	[0.067, 0.337]

B Per-gene held-out performance with bootstrap CIs (Phase 2)

Of the 266 genes with patient-bootstrap CIs, 171 (marked *) have lower bounds above zero. The distribution is long-tailed: a minority of genes are strongly morphology-coupled, most are weakly coupled. This table is generated directly from `results/p2_bootstrap_ci.json` and is not hand-edited.

Table 6: Per-gene held-out M2 correlations (median over held-out sections) with patient-bootstrap 95% CIs, sorted by median r . Genes with CI lower bound > 0 are marked *.

Gene	Symbol	median r	CI lo	CI hi
ENSG00000269028		*	0.450	0.284 0.564
ENSG00000255823		*	0.399	0.237 0.512
ENSG00000169710	FASN*	0.298	0.263	0.321
ENSG00000087460	GNAS*	0.294	0.213	0.363
ENSG00000096384	HSP90AB1*	0.285	0.213	0.348
ENSG00000141736	ERBB2*	0.278	0.137	0.326
ENSG00000132475	H3-3B*	0.262	0.127	0.321
ENSG00000137818	RPLP1*	0.255	0.217	0.340
ENSG00000177600	RPLP2*	0.242	0.148	0.301
ENSG00000184009	ACTG1*	0.238	0.113	0.375
ENSG00000125730	C3*	0.234	0.188	0.292
ENSG00000254709	IGLL5*	0.226	0.050	0.433

continued

	Gene	Symbol	median r	CI lo	CI hi
ENSG00000115414			FN1*	0.223	0.052 0.373
ENSG00000166710			B2M*	0.221	0.138 0.340
ENSG00000108821			COL1A1*	0.220	0.107 0.285
ENSG00000106624			AEBP1*	0.208	0.155 0.263
ENSG00000187514			PTMA*	0.204	0.160 0.336
ENSG00000130203			APOE*	0.191	0.158 0.281
ENSG00000162511			LAPTM5*	0.188	0.083 0.217
ENSG00000204287			HLA-DRA*	0.186	0.130 0.307
ENSG00000234745			HLA-B*	0.180	0.052 0.210
ENSG00000100219			XBPI*	0.177	0.064 0.255
ENSG00000163399			ATP1A1*	0.172	0.095 0.246
ENSG00000026025			VIM*	0.169	0.136 0.234
ENSG00000161960			EIF4A1*	0.166	0.117 0.256
ENSG00000196576			PLXNB2*	0.162	0.027 0.214
ENSG00000164733			CTSB*	0.162	0.116 0.197
ENSG00000180879			SSR4*	0.160	0.059 0.194
ENSG00000164692			COL1A2*	0.158	0.117 0.234
ENSG00000182492			BGN*	0.152	0.082 0.213
ENSG00000149591			TAGLN*	0.146	0.128 0.256
ENSG00000019582			CD74*	0.141	0.109 0.203
ENSG00000142173			COL6A2*	0.139	0.092 0.202
ENSG00000197746			PSAP*	0.137	0.039 0.181
ENSG00000204525			HLA-C*	0.136	0.097 0.162
ENSG00000101439			CST3*	0.135	0.032 0.192
ENSG00000087086			FTL*	0.133	0.072 0.216
ENSG00000251562			MALAT1*	0.133	0.034 0.182
ENSG00000173372			C1QA*	0.130	0.057 0.211
ENSG00000106211			HSPB1*	0.127	0.093 0.148
ENSG00000233276			GPX1	0.126	-0.023 0.173
ENSG00000142949			PTPRF*	0.124	0.064 0.203
ENSG00000197756			RPL37A*	0.123	0.052 0.140
ENSG00000070756			PABPC1*	0.123	0.061 0.148
ENSG00000075624			ACTB*	0.120	0.057 0.154
ENSG00000101335			MYL9*	0.120	0.055 0.165
ENSG00000171345			KRT19*	0.119	0.032 0.179
ENSG00000205542			TMSB4X*	0.118	0.051 0.227
ENSG00000213741			RPS29*	0.118	0.053 0.145
ENSG00000206503			HLA-A*	0.117	0.069 0.148
ENSG00000231500			RPS18*	0.115	0.056 0.188
ENSG00000117450			PRDX1*	0.113	0.062 0.158
ENSG00000108107			RPL28*	0.113	0.053 0.162
ENSG00000159840			ZYX*	0.112	0.049 0.131
ENSG00000089157			RPLP0*	0.112	0.092 0.157
ENSG00000170421			KRT8*	0.112	0.075 0.170

continued

	Gene	Symbol	median r	CI lo	CI hi
ENSG00000133112			TPT1*	0.110	0.018 0.203
ENSG00000177954			RPS27*	0.110	0.063 0.200
ENSG00000130208			APOC1*	0.109	0.052 0.149
ENSG00000117984			CTSD*	0.106	0.028 0.277
ENSG00000163682			RPL9*	0.105	0.063 0.178
ENSG00000142541			RPL13A*	0.104	0.049 0.167
ENSG00000164587			RPS14*	0.100	0.072 0.182
ENSG00000115884			SDC1*	0.099	0.016 0.120
ENSG00000166441			RPL27A*	0.099	0.059 0.191
ENSG00000105223			PLD3*	0.097	0.048 0.110
ENSG00000111640			GAPDH*	0.094	0.024 0.158
__ambiguous[ENSG00000277957+ENSG00000161960]			*	0.093	0.056 0.152
ENSG00000123416			TUBA1B*	0.092	0.034 0.260
ENSG00000112306			RPS12*	0.091	0.063 0.116
ENSG00000124664			SPDEF*	0.089	0.060 0.127
ENSG00000172531			PPP1CA*	0.087	0.066 0.121
ENSG00000163191			S100A11*	0.086	0.038 0.131
ENSG00000279483			*	0.086	0.060 0.108
ENSG00000170889			RPS9*	0.085	0.011 0.125
__ambiguous[ENSG00000241343+ENSG00000257529]			*	0.085	0.041 0.169
ENSG00000143947			RPS27A*	0.085	0.047 0.147
ENSG00000231925			TAPBP*	0.084	0.031 0.119
ENSG00000141741			MIEN1*	0.084	0.034 0.099
ENSG00000108344			PSMD3*	0.083	0.038 0.168
ENSG00000114353			GNAI2*	0.083	0.050 0.101
ENSG00000110651			CD81*	0.083	0.024 0.104
ENSG00000145425			RPS3A*	0.083	0.025 0.105
ENSG00000107223			EDF1*	0.082	0.052 0.109
ENSG00000171863			RPS7	0.081	-0.011 0.133
ENSG00000167526			RPL13*	0.079	0.055 0.190
ENSG00000184897			H1-10*	0.078	0.027 0.108
ENSG00000034510			TMSB10*	0.078	0.031 0.123
ENSG00000109475			RPL34*	0.078	0.056 0.169
ENSG00000167995			BEST1	0.078	-0.001 0.130
ENSG00000182809			CRIP2*	0.078	0.029 0.123
ENSG00000221983			UBA52*	0.077	0.049 0.122
ENSG00000071082			RPL31*	0.076	0.043 0.141
ENSG00000108639			SYNGR2*	0.076	0.038 0.139
__ambiguous[ENSG00000185883+ENSG00000260272]			*	0.076	0.048 0.108
ENSG00000131748			STARD3*	0.076	0.023 0.101
ENSG00000186468			RPS23*	0.076	0.025 0.130
ENSG00000174748			RPL15*	0.076	0.033 0.136
ENSG00000118181			RPS25*	0.076	0.028 0.137
ENSG00000105640			RPL18A*	0.073	0.050 0.124

continued

	Gene	Symbol	median r	CI lo	CI hi
ENSG00000102265			TIMP1*	0.072	0.031 0.120
ENSG00000124172			ATP5F1E*	0.070	0.034 0.195
ENSG000000083845			RPS5*	0.069	0.013 0.095
ENSG000000063177			RPL18*	0.069	0.034 0.081
ENSG00000130520			LSM4*	0.069	0.025 0.122
ENSG00000160789			LMNA*	0.069	0.034 0.117
ENSG00000105193			RPS16*	0.069	0.008 0.102
ENSG00000058262			SEC61A1*	0.069	0.049 0.114
ENSG000000008988			RPS20*	0.068	0.045 0.108
ENSG00000113140			SPARC*	0.068	0.023 0.113
ENSG00000130255			RPL36*	0.068	0.019 0.160
ENSG00000105372			RPS19*	0.068	0.026 0.164
ENSG00000196405			EVL	0.068	-0.006 0.147
ENSG00000138326			RPS24*	0.067	0.034 0.151
ENSG00000179218			CALR*	0.067	0.021 0.088
ENSG00000108518			PFN1*	0.067	0.039 0.110
ENSG00000142669			SH3BGRL3*	0.067	0.018 0.099
ENSG00000224389			C4B*	0.065	0.031 0.132
ENSG00000279274			*	0.065	0.017 0.100
ENSG000000025708			TYMP*	0.065	0.041 0.100
ENSG00000198034			RPS4X	0.064	-0.040 0.103
ENSG00000108679			LGALS3BP*	0.064	0.030 0.093
ENSG00000149806			FAU*	0.064	0.047 0.111
ENSG00000233927			RPS28*	0.064	0.012 0.087
ENSG00000143420			ENSA*	0.063	0.037 0.111
ENSG00000182899			RPL35A*	0.063	0.033 0.092
ENSG00000145592			RPL37*	0.063	0.025 0.117
ENSG00000173369			C1QB*	0.062	0.035 0.105
ENSG00000165215			CLDN3*	0.062	0.030 0.104
ENSG00000147403			RPL10*	0.062	0.019 0.089
ENSG00000164919			COX6C*	0.061	0.039 0.122
ENSG00000272196			*	0.061	0.032 0.107
ENSG00000142534			RPS11*	0.061	0.008 0.100
ENSG00000167468			GPX4*	0.061	0.046 0.082
ENSG000000051523			CYBA*	0.060	0.023 0.095
ENSG00000137154			RPS6*	0.060	0.031 0.123
ENSG00000182774			RPS17*	0.060	0.031 0.096
ENSG00000168528			SERINC2*	0.059	0.046 0.096
ENSG00000173801			JUP*	0.058	0.013 0.095
ENSG000000011304			PTBP1*	0.058	0.017 0.119
ENSG00000131469			RPL27	0.057	-0.008 0.083
ENSG00000106153			CHCHD2	0.057	-0.008 0.146
ENSG00000170315			UBB*	0.057	0.012 0.113
ENSG00000126267			COX6B1*	0.057	0.015 0.088

continued

	Gene	Symbol	median r	CI lo	CI hi
__ambiguous[ENSG00000219200+ENSG00000161939]	*		0.056	0.030	0.094
ENSG00000109062		NHERF1*	0.056	0.014	0.111
ENSG00000130303		BST2*	0.055	0.034	0.088
__ambiguous[ENSG00000254772+ENSG00000255508]			0.055	-0.011	0.074
ENSG00000125970		RALY*	0.055	0.009	0.138
ENSG00000155366		RHOC*	0.054	0.038	0.072
ENSG00000114867		EIF4G1*	0.054	0.022	0.074
ENSG00000156482		RPL30*	0.053	0.008	0.083
ENSG00000196924		FLNA*	0.053	0.043	0.095
ENSG00000142937		RPS8	0.053	-0.035	0.089
ENSG00000141753		IGFBP4	0.053	-0.005	0.107
ENSG00000185624		P4HB*	0.052	0.036	0.138
ENSG00000104964		TLE5*	0.052	0.030	0.064
ENSG00000104529		EEF1D	0.052	-0.019	0.067
ENSG00000172354		GNB2*	0.052	0.021	0.076
ENSG00000147684		NDUFB9*	0.050	0.020	0.082
ENSG00000134419		RPS15A	0.049	-0.002	0.165
ENSG00000132507		EIF5A	0.048	-0.013	0.070
ENSG00000150991		UBC	0.047	-0.013	0.089
ENSG00000142676		RPL11*	0.047	0.004	0.072
ENSG00000052344		PRSS8*	0.046	0.006	0.118
ENSG00000100316		RPL3*	0.046	0.026	0.110
ENSG00000100300		TSPO	0.046	-0.010	0.064
ENSG00000115268		RPS15*	0.045	0.010	0.066
ENSG00000168542		COL3A1	0.045	-0.014	0.253
ENSG00000104904		OAZ1	0.045	-0.007	0.086
ENSG00000174444		RPL4*	0.044	0.031	0.082
ENSG00000160932		LY6E*	0.044	0.032	0.092
ENSG00000213988		ZNF90*	0.042	0.015	0.074
ENSG00000117410		ATP6V0B	0.041	-0.015	0.070
ENSG00000114391		RPL24*	0.041	0.015	0.051
ENSG00000130725		UBE2M*	0.041	0.021	0.081
ENSG00000197958		RPL12	0.039	-0.014	0.072
__ambiguous[ENSG00000124614+ENSG00000270800]			0.039	-0.019	0.062
ENSG00000147604		RPL7	0.039	-0.021	0.091
__ambiguous[ENSG00000257341+ENSG00000213145]	*		0.038	0.005	0.068
ENSG00000172270		BSG*	0.037	0.021	0.049
ENSG00000167815		PRDX2*	0.037	0.019	0.062
__ambiguous[ENSG00000251357+ENSG00000240972]	*		0.036	0.005	0.073
ENSG00000167996		FTH1	0.035	-0.043	0.098
ENSG00000178719		GRINA	0.034	-0.001	0.086
ENSG00000186010		NDUFA13	0.034	-0.031	0.049
ENSG00000163479		SSR2	0.034	-0.008	0.049
ENSG00000135404		CD63	0.033	-0.016	0.063

continued

	Gene	Symbol	median r	CI lo	CI hi
ENSG00000092841		MYL6	0.033	-0.005	0.051
ENSG00000173272		MZT2A	0.032	-0.012	0.058
ENSG00000162244		RPL29	0.032	-0.004	0.075
ENSG00000178980		SELENOW*	0.032	0.004	0.071
ENSG00000105404		RABAC1*	0.032	0.005	0.080
ENSG00000172809		RPL38*	0.031	0.015	0.069
ENSG00000135480		KRT7	0.031	-0.001	0.096
ENSG00000140264		SERF2	0.031	-0.009	0.053
ENSG00000135940		COX5B	0.031	-0.016	0.071
ENSG00000124614		RPS10	0.030	-0.010	0.071
ENSG00000149925		ALDOA	0.030	-0.010	0.065
ENSG00000169223		LMAN2	0.030	-0.000	0.122
ENSG00000149273		RPS3	0.030	-0.010	0.090
ENSG00000115457		IGFBP2	0.029	-0.010	0.095
ENSG00000071553		ATP6AP1*	0.029	0.010	0.084
ENSG00000143774		GUK1	0.028	-0.004	0.072
ENSG00000111669		TPI1	0.028	-0.022	0.071
ENSG00000176046		NUPR1	0.026	-0.010	0.044
ENSG00000171858		RPS21	0.025	-0.015	0.050
ENSG00000214530		STARD10	0.022	-0.020	0.038
ENSG00000140988		RPS2	0.022	-0.002	0.039
ENSG00000136942		RPL35	0.022	-0.035	0.116
ENSG00000152082		MZT2B	0.022	-0.000	0.041
ENSG00000197956		S100A6	0.021	-0.023	0.063
ENSG00000123349		PFDN5*	0.020	0.009	0.052
ENSG00000127540		UQCR11	0.020	-0.023	0.035
ENSG00000204592		HLA-E*	0.019	0.006	0.128
__ambiguous[ENSG00000243678+ENSG00000011052]			0.019	-0.008	0.116
ENSG00000176340		COX8A*	0.019	0.005	0.045
ENSG00000188846		RPL14	0.018	-0.022	0.030
ENSG00000174886		NDUFA11*	0.018	0.011	0.033
ENSG00000167658		EEF2	0.018	-0.015	0.051
ENSG00000171222		SCAND1	0.017	-0.001	0.054
ENSG00000144426		NBEAL1	0.017	-0.009	0.049
ENSG00000148303		RPL7A	0.017	-0.016	0.056
ENSG00000169100		SLC25A6	0.016	-0.005	0.041
ENSG00000145494		NDUFS6	0.016	-0.008	0.045
ENSG00000111057		KRT18	0.016	-0.010	0.033
ENSG00000158710		TAGLN2*	0.016	0.002	0.049
__ambiguous[ENSG00000185885+ENSG00000185201]			0.015	-0.047	0.054
ENSG00000030582		GRN	0.015	-0.010	0.094
ENSG00000277791		PSMB3	0.014	-0.030	0.084
ENSG00000123989		CHPF	0.012	-0.011	0.091
ENSG00000131143		COX4I1	0.011	-0.016	0.022

continued

	Gene	Symbol	median r	CI lo	CI hi
ENSG00000198242		RPL23A	0.010	-0.037	0.045
ENSG00000161016		RPL8	0.010	-0.012	0.085
ENSG00000185825		BCAP31	0.009	-0.012	0.031
ENSG00000141522		ARHGDIA	0.008	-0.015	0.038
ENSG00000277972		CISD3	0.008	-0.013	0.043
...ambiguous[ENSG00000262526+ENSG00000170296]			0.007	-0.043	0.051
ENSG00000185499		MUC1	0.005	-0.076	0.029
ENSG00000074800		ENO1	0.005	-0.052	0.051
ENSG00000172757		CFL1	0.004	-0.020	0.029
ENSG00000177700		POLR2L	0.003	-0.025	0.037
ENSG00000178952		TUFM	0.003	-0.019	0.068
ENSG00000165949		IFI27	0.001	-0.019	0.017
ENSG00000163220		S100A9	0.001	-0.028	0.050
ENSG00000187608		ISG15	0.001	-0.021	0.067
ENSG00000126247		CAPNS1	0.000	-0.022	0.013
ENSG00000111341		MGP	-0.002	-0.099	0.050
ENSG00000110492		MDK	-0.002	-0.046	0.038
ENSG00000135390		ATP5MC2	-0.002	-0.056	0.038
...ambiguous[ENSG00000273259+ENSG00000196136]			-0.004	-0.035	0.018
ENSG00000140990		NDUFB10	-0.004	-0.019	0.023
ENSG00000103363		ELOB	-0.005	-0.029	0.034
ENSG00000144713		RPL32	-0.006	-0.047	0.023
ENSG00000184840		TMED9	-0.006	-0.093	0.035
ENSG00000143546		S100A8	-0.006	-0.054	0.047
ENSG00000204628		RACK1	-0.008	-0.020	0.046
ENSG00000142089		IFITM3	-0.011	-0.041	0.046
ENSG00000160862		AZGP1	-0.011	-0.066	0.005
ENSG00000246705		H2AJ	-0.012	-0.040	0.039
ENSG00000159335		PTMS	-0.020	-0.030	0.027
ENSG00000160180		TFF3	-0.023	-0.079	0.013
ENSG00000125691		RPL23	-0.043	-0.099	0.004
ENSG00000110092		CCND1	-0.046	-0.119	0.054
ENSG00000125534		PPDPF	-0.050	-0.071	0.007
ENSG00000108298		RPL19	-0.065	-0.137	-0.012

C Locked formulas index

1. Pearson $r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}}$
2. Spearman ρ (Pearson on ranks)
3. Ridge $\hat{\beta} = (X^\top X + \lambda P)^{-1} X^\top Y$
4. Gaussian kernel $W_{ij} = \exp(-d_{ij}^2 / 2\ell^2)$, row-normalized
5. Spatial smoothing $\hat{y}^{(2)} = (1 - \alpha)\hat{y} + \alpha W \hat{y}$

6. Moran's I (see Methods)
7. Beer–Lambert $OD = -\ln(I/I_0)$; deconvolution $C = ODM^{-1}$
8. GLCM contrast / homogeneity / energy / correlation (see Methods)
9. Shannon entropy $H = -\sum p \log_2 p$
10. AUC via Mann–Whitney $U/(n_1 n_0)$
11. Benjamini–Hochberg adjusted $p_{(i)} \cdot m/i$
12. Permutation $p = (1 + \#\{|null| \geq |obs|\})/(1 + n)$
13. Wilcoxon signed-rank statistic
14. Brier score $\frac{1}{n} \sum (p_i - y_i)^2$
15. Softmax $\alpha_i = e^{s_i/\tau} / \sum_j e^{s_j/\tau}$
16. Library normalization $\log(1 + 10^4 x / \sum x)$

Provenance note. The appendix tables from here on (per-fold, transport, scanner coefficients, top predictable genes, enrichment, ledgers, cohort manifest) are Phase-1 artifacts from the handcrafted-feature pipeline (commit 84b076a), preserved for provenance. Phase-2 equivalents are Appendix ??–?? and the refreshed scanner outputs in `results/`.

D Per-fold results

Patient	sections	λ	α	M1 median r	M2 median r
BC23287	3	100.0	0.5	-0.007	-0.010
BC23450	3	100.0	0.4	0.016	0.019
BC23567	3	100.0	0.5	0.009	0.009
BC23810	3	100.0	0.5	0.011	0.014
BC23901	2	100.0	0.5	0.009	0.013
BC23903	3	100.0	0.5	0.043	0.055
BC23944	3	100.0	0.5	0.029	0.033
BC24044	3	100.0	0.5	0.040	0.049
BC24105	3	100.0	0.5	-0.017	-0.022
BC24220	3	100.0	0.5	0.025	0.034

E Transport results

Subtype	Section	median per-gene r
Luminal_A	BC23268_C1	0.072
Luminal_A	BC23268_C2	0.061

Subtype		Section	median r
Luminal_A	BC23268_D1	0.076	
Luminal_A	BC23269_C1	0.040	
Luminal_A	BC23269_C2	0.025	
Luminal_A	BC23269_D1	0.031	
Luminal_A	BC23272_D2	0.050	
Luminal_A	BC23272_E1	0.075	
Luminal_A	BC23272_E2	0.021	
Luminal_A	BC24223_D2	-0.024	
Luminal_A	BC24223_E1	-0.013	
Luminal_A	BC24223_E2	0.000	
Luminal_B	BC23270_D2	-0.007	
Luminal_B	BC23270_E1	0.041	
Luminal_B	BC23270_E2	0.023	
Luminal_B	BC23277_D2	0.015	
Luminal_B	BC23277_E1	0.027	
Luminal_B	BC23277_E2	-0.025	
Luminal_B	BC23506_C1	0.016	
Luminal_B	BC23506_C2	-0.004	
Luminal_B	BC23506_D1	-0.001	
Luminal_B	BC23508_D2	0.037	
Luminal_B	BC23508_E1	0.026	
Luminal_B	BC23508_E2	0.043	
Luminal_B	BC23895_C1	0.013	
Luminal_B	BC23895_C2	0.014	
Luminal_B	BC23895_D1	0.005	
TNBC	BC23209_C1	0.004	
TNBC	BC23209_C2	0.004	
TNBC	BC23209_D1	-0.024	
TNBC	BC23288_D2	0.021	
TNBC	BC23288_E1	0.051	
TNBC	BC23288_E2	0.029	
TNBC	BC23377_C1	0.010	
TNBC	BC23377_C2	-0.005	
TNBC	BC23377_D1	-0.007	
TNBC	BC23803_D2	0.006	
TNBC	BC23803_E1	0.038	
TNBC	BC23803_E2	0.019	

F Scanner coefficients

Feature	coefficient	z	p
glcm0_energy	24.566	5.52	3.41e-08
glcm1_energy	-23.145	-5.13	2.90e-07

Feature	coef	z	p
glcm3_energy	12.963	3.56	3.67e-04
glcm2_energy	-12.931	-3.60	3.23e-04
b_mean	11.592	18.82	5.46e-79
r_mean	-5.979	-19.25	1.51e-82
eosin_mean	5.229	12.23	2.26e-34
dab_mean	4.647	15.01	6.58e-51
r_std	2.601	12.53	5.15e-36
g_mean	-2.417	-7.49	6.93e-14
hema_mean	2.409	10.49	9.52e-26
hema_std	-2.103	-17.03	4.88e-65
eosin_std	2.020	12.40	2.58e-35
b_std	-1.372	-8.12	4.63e-16
glcm2_homog	-1.319	-1.50	1.33e-01
g_std	-1.137	-5.40	6.84e-08
glcm1_homog	-0.915	-1.10	2.69e-01
gray_entropy	-0.745	-9.48	2.61e-21
glcm3_homog	0.649	0.75	4.55e-01
glcm0_homog	-0.617	-0.74	4.60e-01
glcm1_contrast	0.528	0.96	3.35e-01
r_skew	-0.515	-3.55	3.84e-04
hema_entropy	0.422	11.23	2.77e-29
glcm3_corr	0.387	4.08	4.59e-05
glcm1_corr	-0.365	-4.66	3.18e-06
grad_mag_mean	-0.365	-2.00	4.55e-02
dab_std	0.279	3.92	8.80e-05
glcm2_contrast	-0.247	-0.43	6.66e-01
g_skew	0.236	1.53	1.25e-01
glcm0_corr	-0.213	-2.69	7.05e-03
glcm0_contrast	0.189	0.34	7.34e-01
glcm2_corr	-0.135	-1.44	1.51e-01
hema_dense_frac	0.133	5.05	4.52e-07
r_kurt	-0.114	-0.76	4.48e-01
hema_fracpos	0.078	3.38	7.22e-04
b_kurt	-0.065	-0.42	6.77e-01
dab_fracpos	0.045	2.01	4.47e-02
glcm3_contrast	0.019	0.03	9.74e-01
b_skew	0.015	0.10	9.19e-01
eosin_fracpos	-0.006	-0.24	8.09e-01
g_kurt	0.001	0.01	9.95e-01

G Top predictable genes

Gene (Ensembl)	M1 median r	M2 median r
ENSG00000177600	0.144	0.154
ENSG00000096384	0.140	0.138
ENSG00000141738	0.115	0.140
ENSG00000137818	0.111	0.144
ENSG00000269028	0.110	0.153
ENSG00000108821	0.109	0.137
ENSG00000132475	0.105	0.140
ENSG00000255823	0.104	0.122
ENSG00000166710	0.098	0.114
ENSG00000087086	0.090	0.104
ENSG00000196576	0.087	0.089
ENSG00000279483	0.087	0.104
ENSG00000130203	0.086	0.120
ENSG00000161960	0.078	0.116
ENSG00000168528	0.076	0.095
ENSG00000115414	0.075	0.082
ENSG00000169710	0.073	0.090
ENSG00000141736	0.068	0.077
ENSG00000087460	0.068	0.092
ENSG00000234745	0.067	0.080
ENSG00000058262	0.064	0.082
ENSG00000102265	0.063	0.054
ENSG00000204287	0.061	0.079
ENSG00000177954	0.061	0.070
ENSG00000075624	0.061	0.085
ENSG00000108344	0.060	0.052
ENSG00000170421	0.059	0.061
ENSG00000089157	0.056	0.058
ENSG00000108518	0.055	0.070
ENSG00000132507	0.055	0.061
ENSG00000130255	0.054	0.073
ENSG00000205542	0.053	0.069
ENSG00000164692	0.053	0.065
ENSG00000197746	0.053	0.069
ENSG00000026025	0.052	0.069
ENSG00000083845	0.051	0.051
ENSG00000100097	0.051	0.055
ENSG00000182899	0.050	0.050
ENSG00000167526	0.049	0.045
ENSG00000142937	0.048	0.046
ENSG00000130725	0.048	0.047
ENSG00000163399	0.047	0.069
ENSG00000164733	0.047	0.064
ENSG00000197756	0.046	0.060
ENSG00000100219	0.046	0.047

Gene	M1	M2
ENSG00000108107	0.046	0.050
ENSG00000108639	0.046	0.038
ENSG00000105223	0.046	0.052
__ambiguous [ENSG00000277957+ENSG00000161960]	0.045	0.071
ENSG00000134419	0.044	0.065
ENSG00000019582	0.043	0.038
ENSG00000149806	0.043	0.037
ENSG00000159840	0.043	-0.006
ENSG00000142949	0.043	0.053
ENSG00000198034	0.042	0.031
ENSG00000184009	0.041	0.042
ENSG00000105193	0.041	0.047
__ambiguous [ENSG00000185883+ENSG00000260272]	0.041	0.050
ENSG00000125970	0.040	0.069
ENSG00000164587	0.040	0.049
ENSG00000158710	0.040	0.031
ENSG00000125730	0.039	0.043
ENSG00000123416	0.038	0.044
ENSG00000173369	0.038	0.034
ENSG00000142173	0.037	0.049
ENSG00000114867	0.037	0.045
ENSG00000173272	0.035	0.023
ENSG00000117450	0.035	0.062
ENSG00000279274	0.035	0.054
ENSG00000229117	0.035	0.026
ENSG00000231500	0.034	0.047
ENSG00000130520	0.034	0.031
ENSG00000106153	0.034	0.023
ENSG00000142541	0.034	0.041
ENSG00000163191	0.034	0.051
ENSG00000231925	0.033	0.037
ENSG00000145425	0.033	0.050
ENSG00000104904	0.033	0.038
ENSG00000131748	0.033	0.056
ENSG00000105404	0.032	0.028
__ambiguous [ENSG00000251357+ENSG00000240972]	0.032	0.024
ENSG00000196405	0.032	0.032
ENSG00000126267	0.031	0.039
ENSG00000174444	0.030	0.023
ENSG00000185624	0.030	0.028
ENSG00000118181	0.029	0.030
ENSG00000111640	0.029	0.033
ENSG00000187514	0.028	0.056
ENSG00000170315	0.028	0.015
ENSG00000224389	0.028	0.019

	Gene	M1	M2
ENSG00000233927		0.028	0.022
ENSG00000143420		0.027	0.037
ENSG00000213741		0.027	0.023
ENSG00000051523		0.027	0.008
ENSG00000277791		0.027	0.042
ENSG00000071082		0.027	0.051
ENSG00000128322		0.027	0.003
ENSG00000071553		0.026	0.038
ENSG00000277972		0.026	0.016
ENSG00000163682		0.025	0.043
ENSG00000109062		0.025	0.040
ENSG00000272196		0.025	0.025
ENSG00000034510		0.024	0.014
ENSG00000123989		0.024	0.020
ENSG00000113140		0.024	0.022
__ambiguous [ENSG00000243678+ENSG00000011052]		0.023	0.029
ENSG00000171863		0.023	0.042
ENSG00000140264		0.023	0.034
ENSG00000167468		0.022	0.026
ENSG00000178952		0.022	0.017
__ambiguous [ENSG00000241343+ENSG00000257529]		0.022	0.024
ENSG00000204525		0.022	0.056
ENSG00000141741		0.022	0.010
ENSG00000170889		0.021	0.019
ENSG00000141522		0.021	0.025
ENSG00000142676		0.021	0.015
ENSG00000140988		0.021	0.020
ENSG00000110651		0.020	0.026
ENSG00000115457		0.020	0.004
ENSG00000162244		0.020	0.015

H Dataset ledger composition

The 629-record accession ledger summarized by source, with example accessions; the full ledger is `results/accession_ledger.csv`.

Source	Records	Level	Example accession
CELLxGENE	40	metadata-mined	cxg:af893e86-8e9f-41f1-a474-ef05359b1fb7; cxg:3a5dbf8a-9b3e-4309-b4c5-d8a024f83734; cxg:16876983-d454-43db-a8ec-77fafca9bf38
GDC	80	metadata-mined	TCGA-LUAD; ALCHEMIST-ALCH; MATCH-S1
GEO-GSE	400	metadata-mined	GSE344506; GSE348445; GSE337859

Source	Records		Level	Examples
Mendeley- 29ntw7sh4r-v5	69	analyzed-primary	BC23287_C1; BC23287_C2; BC23287_D1	
TCIA	40	metadata-mined	tcia:4D-Lung; tcia:A091105; tcia:ACNS0332	
Total	629			

I Cohort manifest

One row per analyzed section. Tumor fraction is the share of spots under the pathologist tumor annotation for that section.

	Patient	Section	Subtype	Spots	Tumor frac.	HER2+
BC23287	BC23287_C1		HER2_luminal	256	0.00	yes
BC23287	BC23287_C2		HER2_luminal	334	0.00	yes
BC23287	BC23287_D1		HER2_luminal	357	0.00	yes
BC23450	BC23450_D2		HER2_luminal	316	0.00	yes
BC23450	BC23450_E1		HER2_luminal	301	0.00	yes
BC23450	BC23450_E2		HER2_luminal	318	0.00	yes
BC23901	BC23901_C2		HER2_luminal	292	0.00	yes
BC23901	BC23901_D1		HER2_luminal	297	0.00	yes
BC23944	BC23944_D2		HER2_luminal	414	0.00	yes
BC23944	BC23944_E1		HER2_luminal	423	0.00	yes
BC23944	BC23944_E2		HER2_luminal	403	0.00	yes
BC24220	BC24220_D2		HER2_luminal	432	0.00	yes
BC24220	BC24220_E1		HER2_luminal	428	0.00	yes
BC24220	BC24220_E2		HER2_luminal	437	0.00	yes
BC23567	BC23567_D2		HER2_non_luminal	587	0.00	yes
BC23567	BC23567_E1		HER2_non_luminal	572	0.00	yes
BC23567	BC23567_E2		HER2_non_luminal	570	0.00	yes
BC23810	BC23810_D2		HER2_non_luminal	691	0.00	yes
BC23810	BC23810_E1		HER2_non_luminal	695	0.00	yes
BC23810	BC23810_E2		HER2_non_luminal	713	0.00	yes
BC23903	BC23903_C1		HER2_non_luminal	441	0.00	yes
BC23903	BC23903_C2		HER2_non_luminal	468	0.00	yes
BC23903	BC23903_D1		HER2_non_luminal	464	0.00	yes
BC24044	BC24044_D2		HER2_non_luminal	614	0.00	yes
BC24044	BC24044_E1		HER2_non_luminal	603	0.00	yes
BC24044	BC24044_E2		HER2_non_luminal	511	0.00	yes
BC24105	BC24105_C1		HER2_non_luminal	325	0.00	yes
BC24105	BC24105_C2		HER2_non_luminal	297	0.00	yes
BC24105	BC24105_D1		HER2_non_luminal	338	0.00	yes
BC23268	BC23268_C1		Luminal_A	435	0.00	no
BC23268	BC23268_C2		Luminal_A	498	0.00	no
BC23268	BC23268_D1		Luminal_A	492	0.00	no
BC23269	BC23269_C1		Luminal_A	434	0.00	no

	Patient	Section	Subtype	Spots	Tumor	HER2+	
BC23269	BC23269_C2		Luminal_A	435	0.00	no	
BC23269	BC23269_D1		Luminal_A	433	0.00	no	
BC23272	BC23272_D2		Luminal_A	565	0.00	no	
BC23272	BC23272_E1		Luminal_A	578	0.00	no	
BC23272	BC23272_E2		Luminal_A	548	0.00	no	
BC24223	BC24223_D2		Luminal_A	368	0.00	no	
BC24223	BC24223_E1		Luminal_A	362	0.00	no	
BC24223	BC24223_E2		Luminal_A	363	0.00	no	
BC23270	BC23270_D2		Luminal_B	284	0.00	no	
BC23270	BC23270_E1		Luminal_B	283	0.00	no	
BC23270	BC23270_E2		Luminal_B	284	0.00	no	
BC23277	BC23277_D2		Luminal_B	506	0.00	no	
BC23277	BC23277_E1		Luminal_B	513	0.00	no	
BC23277	BC23277_E2		Luminal_B	566	0.00	no	
BC23506	BC23506_C1		Luminal_B	489	0.00	no	
BC23506	BC23506_C2		Luminal_B	502	0.00	no	
BC23506	BC23506_D1		Luminal_B	482	0.00	no	
BC23508	BC23508_D2		Luminal_B	478	0.00	no	
BC23508	BC23508_E1		Luminal_B	473	0.00	no	
BC23508	BC23508_E2		Luminal_B	504	0.00	no	
BC23895	BC23895_C1		Luminal_B	498	0.00	no	
BC23895	BC23895_C2		Luminal_B	454	0.00	no	
BC23895	BC23895_D1		Luminal_B	518	0.00	no	
BC23209	BC23209_C1		TNBC	294	0.00	no	
BC23209	BC23209_C2		TNBC	318	0.00	no	
BC23209	BC23209_D1		TNBC	331	0.00	no	
BC23288	BC23288_D2		TNBC	464	0.00	no	
BC23288	BC23288_E1		TNBC	480	0.00	no	
BC23288	BC23288_E2		TNBC	478	0.00	no	
BC23377	BC23377_C1		TNBC	597	0.00	no	
BC23377	BC23377_C2		TNBC	687	0.00	no	
BC23377	BC23377_D1		TNBC	601	0.00	no	
BC23803	BC23803_D2		TNBC	402	0.00	no	
BC23803	BC23803_E1		TNBC	340	0.00	no	
BC23803	BC23803_E2		TNBC	421	0.00	no	

J Pathway enrichment of the top-25 predictable genes

g:Profiler g:GOST (GO:BP, KEGG, Reactome) on the 25 genes with the highest held-out M1 correlation; full response in results/enrichment_top25.json.

	Source	Term	<i>p</i>	Genes
KEGG		Antigen processing and presentation	9.22e-04	23
REAC		Scavenging by Class A Receptors	1.82e-03	23

	Source	Term	<i>p</i>	Genes
GO:BP		antigen processing and presentation of endogenous peptide antigen	3.12e-03	23
REAC		GRB7 events in ERBB2 signaling	6.68e-03	23
GO:BP		antigen processing and presentation of endogenous antigen	6.90e-03	23
GO:BP		wound healing	7.13e-03	23
GO:BP		protein biosynthetic process	8.61e-03	23
GO:BP		translation	8.61e-03	23
REAC		Disease	9.90e-03	23
KEGG		Vibrio cholerae infection	1.04e-02	23
GO:BP		positive regulation of T cell mediated cytotoxicity	1.19e-02	23
REAC		Axon guidance	1.37e-02	23
GO:BP		regulation of cell adhesion	1.57e-02	23
GO:BP		regulation of developmental process	1.61e-02	23
KEGG		Phagosome	1.68e-02	23

K Test inventory

Test module	Tests
features	11
metrics	21
Total	32

The 32 unit tests pin every metric against known values (perfect/anti correlation, constant-input degeneracies, ridge recovery and shrinkage, kernel normalization, smoothing identity at $\alpha = 0$, Moran's I sign on clustered vs alternating patterns, AUC 0/0.5/1 anchors, BH monotonicity, permutation- p bounds, Brier anchors, softmax normalization and temperature behavior) and every feature extractor against physical expectations (white-field OD ≈ 0 , constant-image GLCM energy 1 and contrast 0, uniform-histogram entropy maximum, feature-vector finiteness and name uniqueness, textured-vs-flat discriminability).

L Phase-2 scanner: per-section performance

Table 15: Phase-2 scanner: per-section patient-held-out AUC. Sections with undefined AUC (tumor fraction 1.0: BC24105_C1, BC24105_C2) are excluded; BC24105_D1 is retained and reported as a named outlier.

Patient	Section	<i>n</i> spots	Tumor frac.	AUC
BC23287	BC23287_C1	256	0.312	0.992
BC23287	BC23287_C2	334	0.266	0.990
BC23287	BC23287_D1	357	0.258	0.667
BC23450	BC23450_D2	316	0.883	0.885

Patient	Section	<i>n</i> spots	Tumor frac.	AUC
BC23450	BC23450_E1	301	0.934	0.887
BC23450	BC23450_E2	318	0.909	0.912
BC23567	BC23567_D2	587	0.472	0.845
BC23567	BC23567_E1	572	0.497	0.841
BC23567	BC23567_E2	570	0.488	0.845
BC23810	BC23810_D2	691	0.889	0.926
BC23810	BC23810_E1	695	0.883	0.952
BC23810	BC23810_E2	713	0.871	0.936
BC23901	BC23901_C2	292	0.442	0.826
BC23901	BC23901_D1	297	0.529	0.849
BC23903	BC23903_C1	441	0.213	0.921
BC23903	BC23903_C2	468	0.171	0.917
BC23903	BC23903_D1	464	0.213	0.932
BC23944	BC23944_D2	414	0.836	0.976
BC23944	BC23944_E1	423	0.823	0.980
BC23944	BC23944_E2	403	0.851	0.993
BC24044	BC24044_D2	614	0.135	0.973
BC24044	BC24044_E1	603	0.134	0.963
BC24044	BC24044_E2	511	0.094	0.972
BC24105	BC24105_C1	325	1.000	—
BC24105	BC24105_C2	297	1.000	—
BC24105	BC24105_D1	338	0.997	0.065
BC24220	BC24220_D2	432	0.884	0.924
BC24220	BC24220_E1	428	0.930	0.954
BC24220	BC24220_E2	437	0.911	0.941

M Phase-2 scanner: subgroup analyses

Table 16: Phase-2 scanner subgroup performance (27 evaluable sections).

Subgroup	<i>n</i>	Median AUC	IQR	Range
HER2 luminal	14	0.933	[0.885, 0.979]	[0.667, 0.993]
HER2 non luminal	13	0.926	[0.845, 0.952]	[0.065, 0.973]
low	9	0.963	[0.921, 0.973]	[0.667, 0.992]
mid	9	0.849	[0.845, 0.976]	[0.826, 0.993]
high	9	0.924	[0.887, 0.941]	[0.065, 0.954]

N Phase-2 scanner: coefficient stability

O Phase-2 scanner: label-permutation controls

Table 17: Phase-2 scanner coefficient stability across the 10 LOPO folds (768-dimensional embedding-logistic coefficient vectors).

Metric	Value
Pairwise Spearman, median	0.857
Pairwise Spearman, bootstrap 95% CI	[0.839, 0.874]
Pairwise Spearman, range	[0.753, 0.942]
Top-50 feature Jaccard, median	0.620
Top-50 sign consistency, median	1.000

Table 18: Phase-2 scanner within-section label-permutation control (20 permutations per section): true AUC vs the permutation null. Exceedances counts permuted fits reaching the true AUC.

Section	True AUC	Null median	Null p95	Exceedances
BC23287_C1	0.992	0.504	0.565	0/20
BC23287_C2	0.990	0.507	0.526	0/20
BC23287_D1	0.667	0.515	0.575	0/20
BC23450_D2	0.885	0.474	0.556	0/20
BC23450_E1	0.887	0.503	0.591	0/20
BC23450_E2	0.912	0.509	0.595	0/20
BC23567_D2	0.845	0.497	0.515	0/20
BC23567_E1	0.841	0.493	0.532	0/20
BC23567_E2	0.845	0.497	0.532	0/20
BC23810_D2	0.926	0.509	0.563	0/20
BC23810_E1	0.952	0.514	0.563	0/20
BC23810_E2	0.936	0.490	0.537	0/20
BC23901_C2	0.826	0.519	0.555	0/20
BC23901_D1	0.849	0.510	0.562	0/20
BC23903_C1	0.921	0.506	0.557	0/20
BC23903_C2	0.917	0.504	0.547	0/20
BC23903_D1	0.932	0.514	0.541	0/20
BC23944_D2	0.976	0.507	0.558	0/20
BC23944_E1	0.980	0.498	0.533	0/20
BC23944_E2	0.993	0.518	0.569	0/20
BC24044_D2	0.973	0.500	0.559	0/20
BC24044_E1	0.963	0.503	0.570	0/20
BC24044_E2	0.972	0.514	0.578	0/20
BC24105_D1	0.065	0.585	0.899	18/20
BC24220_D2	0.924	0.500	0.574	0/20
BC24220_E1	0.954	0.508	0.575	0/20
BC24220_E2	0.941	0.524	0.591	0/20

P Multi-task per-fold results

Table 19: Multi-task legs per held-out patient (median per-gene Pearson over the 250 train-selected genes). ridge = committed independent ridge; refold = same ridge refit by the multi-task harness (sanity); PLS = partial least squares (25 components); RRR = reduced-rank ridge (rank 25); enet = multi-output elastic net (alpha 1e-3, l1-ratio 0.5, max 500 iterations, random selection, seed 20260929; all folds hit the iteration cap - non-convergence declared).

patient	ridge	refold	PLS	RRR	enet	enet n_iter
BC23287	0.0588	0.0520	0.0896	0.0721	0.0467	500
BC23450	0.0388	0.0484	0.0734	0.0562	0.0439	500
BC23567	0.0334	0.0404	0.0491	0.0500	0.0386	500
BC23810	0.0623	0.0720	0.0786	0.0766	0.0658	500
BC23901	0.0417	0.0314	0.0494	0.0390	0.0277	500
BC23903	0.1063	0.1006	0.1077	0.1041	0.0948	500
BC23944	0.0390	0.0323	0.0442	0.0302	0.0268	500
BC24044	0.0853	0.0846	0.0966	0.0910	0.0797	500
BC24105	0.0312	0.0315	0.0400	0.0366	0.0279	500
BC24220	0.0507	0.0542	0.0611	0.0630	0.0511	500